

ATHUL S

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🌐 linkedin.com/in/athul-s-engineer 📅 17th November 2004

SUMMARY

Electronics and Communication Engineering graduate skilled in Image Processing, MATLAB, Machine Learning, and Computer Networks with hands-on experience in real-time embedded and IoT-based projects.

EXPERIENCE

Intern

NeST Technologies

04/2025 – 04/2025

Kochi, Kerala

- Completed 5-day, 30-hour practical training on Arduino/ESP8266-based IoT development.
- Built 2 mini-projects involving temperature and motion sensing, achieving 100% wireless data transmission via Wi-Fi.
- Implemented Histogram Equalization using OpenCV in Python to improve contrast and visibility in low-light images.

Intern

KSEB

01/2025 – 01/2025

Trivandrum, Kerala

- Observed real-time monitoring of 3 major grid substations using SCADA (Supervisory Control and Data Acquisition).
- Gained hands-on exposure to 5+ industrial communication protocols, including Modbus, TCP/IP, and IEC 60870, for reliable real-time grid data transmission and monitoring.
- Analyzed industrial power grid communication workflows to understand real-time SCADA-based monitoring and data transmission systems.

EDUCATION

BTech- Electronics and Communication

Amrita Vishwa Vidyapeetham

CGPA - 7.08

2022 – 2026

Karunagapally, Kollam

Higher Secondary -12th STD

Jyothis Central School

2020 – 2022

Kazhakuttam

10th STD

Blue Mount Public School

2018 – 2020

Thonnakkal

SKILLS

Programming

Python, Assembly Language

Image Processing

Digital Communication

Computer Networks

Ansys, Keil μ Vision, MATLAB, Cisco Packet Tracer, Arduino IDE

PROJECT

Chaakra IoT-Based GPS Bus Tracking System

10/2025 – 05/2026

- Built a real-time bus tracking and booking system using React Native, Firebase, ESP32, and GPS modules.
- Implemented live GPS synchronization, ride management, ETA tracking, and real-time Firebase updates.
- Integrated Leaflet Maps for live bus visualization and optimized backend performance with dynamic Firebase listeners.
- Processed GPS data updates for multiple buses with Firebase synchronization, achieving live tracking refresh intervals under 5 seconds.

FPGA Candy Vending Machine (Basys 3) - Used 6 Verilog modules, including FSM, debounce, BCD converter, and display driver. - Simulated and tested vending logic with 3 coin inputs and 4 item outputs, achieving real-time response <1s. - Implemented and verified FSM-based vending control logic using Verilog simulation tools.	02/2025 – 03/2025
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Compact Automotive Antenna - Designed a 2.5 × 2.5 cm antenna prototype operating at 5.8 GHz for V2V communication. - Achieved >85% radiation efficiency and <10 ms latency in dense vehicular simulation environments. - Simulated antenna performance parameters, including return loss, gain, and radiation pattern, using Ansys HFSS.	11/2024 – 01/2025
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Microcontroller Temperature Data Logger - Logged temperature every 10 seconds with ±0.5°C accuracy over 24-hour cycles using LM35 + ATmega328. - Achieved 100% data integrity and built a 16x2 LCD for live readings and status. - Developed embedded C logic for periodic sensor acquisition and real-time LCD display updates.	12/2024 – 12/2024
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COURSEWORK

Google Advanced Data Analytics Professional Certificate <i>Google Career Certificates, Coursera</i> Completed an 8-course professional certification covering Python, machine learning, data analysis, statistics, predictive modeling, and data visualization.	2026
Machine Learning <i>Amrita Vishwa Vidyapeetham</i> The training consisted of an Introduction to Machine Learning, Data, Python, Data Exploration and Pre-processing, linear regression, and Dimensionality Reduction.	2024

INVOLVEMENT

Old Age Home Program <i>Overall Coordinator</i> - Coordinated social engagement activities for 20+ elderly residents to promote interaction, emotional well-being, and community participation. - Organized and managed event activities in collaboration with the Student Social Responsibility (SSR) initiative. - Assisted in coordinating student volunteers and activity planning for community engagement programs involving 20+ participants.	11/2024 – 11/2024
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